

Engineering Progress Report – CRIMSy

Repositories

- master

<https://github.com/ipb-halle/CRIMSy>

- crimsy_training

https://github.com/ipb-halle/CRIMSy/tree/crimsy_training

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1. General Scope

- Worked on modernization of existing CRIMSy platform (IPB)
- Focus: backend refactoring + API layer + frontend decoupling
- Migrated system toward Jakarta EE and REST-based architecture
- Introduced OpenAPI-driven API contract approach
- Separated frontend (React app) from legacy UI layer

2. Backend (Jakarta EE Migration & Refactoring)

- Migrated javax.* → jakarta.*
- Refactored backend package structure for modularity
- Introduced REST APIs for Authentication, User management, Search functionality
- Implemented OpenAPI specification (user-api.yml)
- Generated API interfaces and models using OpenAPI generator
- Manually implemented service layer based on generated interfaces

3. Authentication & Session Management

- Implemented Bearer token authentication
- Introduced database-backed session storage (user_sessions)
- Implemented login and logout endpoints
- Session expiration mechanism
- Scheduled cleanup job for expired sessions
- CORS filter for frontend communication
- Security filter and interceptor layer (AuthFilter, SecurityInterceptor)

4. Frontend (React + TypeScript – ui-app)

- Created standalone frontend application
- Tech stack: React + TypeScript
- Integrated OpenAPI-generated TypeScript client
- Login / logout flow
- Dashboard page
- User management UI
- Search interface with filters
- Pagination (React DataTable)
- Export functionality (CSV / PDF)
- Role-based UI rendering
- Separated from legacy UI completely
- Uses API-only communication with backend
- Introduced hooks/adapters structure (useAuth, useUsers)

5. User Management

- Implemented paginated user list API
- Role-based access (admin vs user)
- Delete user functionality
- Update user role functionality
- Partial CRUD implementation (create/update in progress)
- Frontend-backend synchronization for user operations

6. Search Functionality

- In-memory dummy data (initial implementation)
- Search UI with filters and advanced panel
- Pagination and result table view
- Backend search API introduced but partially static
- Full persistence layer not yet completed

7. Architecture & Code Structure Changes

- Adopted API-first development (OpenAPI contract)
- Decoupled frontend and backend
- Modular frontend structure (adapters, hooks, services, UI components)
- Reduced duplicated authentication logic
- Centralized API client usage
- Improved session handling consistency

8. Commit Summary

- ~243 commits in branch
- ~125 files changed
- Authentication system redesign
- Frontend application creation
- API restructuring
- Search and user management features
- Session/token lifecycle handling
- UI/UX improvements and refactoring

9. Current State / Limitations

- Search backend partially mock-based
- Full CRUD not fully finalized
- Some legacy UI files still exist
- Further persistence migration required

10. Summary

- System migrated toward REST API architecture
- Jakarta EE compatibility achieved
- Frontend fully decoupled (React)
- Authentication and session system redesigned
- Core functionality preserved and modernized